



SSD PLAYER 开发环境搭建



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{SigmaStar Display}

{Smart Display}

{SSD201 + 1.0}

REVISION HISTORY

Revision No.	Description	Date
0.1	Initial release	06/06/2019

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精简版 FLYTHINGS IDE 环境搭建

1.1. 安装 flythings 工具

到 FTP 如下路径下载 flythings 工具，并安装，具体使用方法参考:IDE 使用参考.pdf

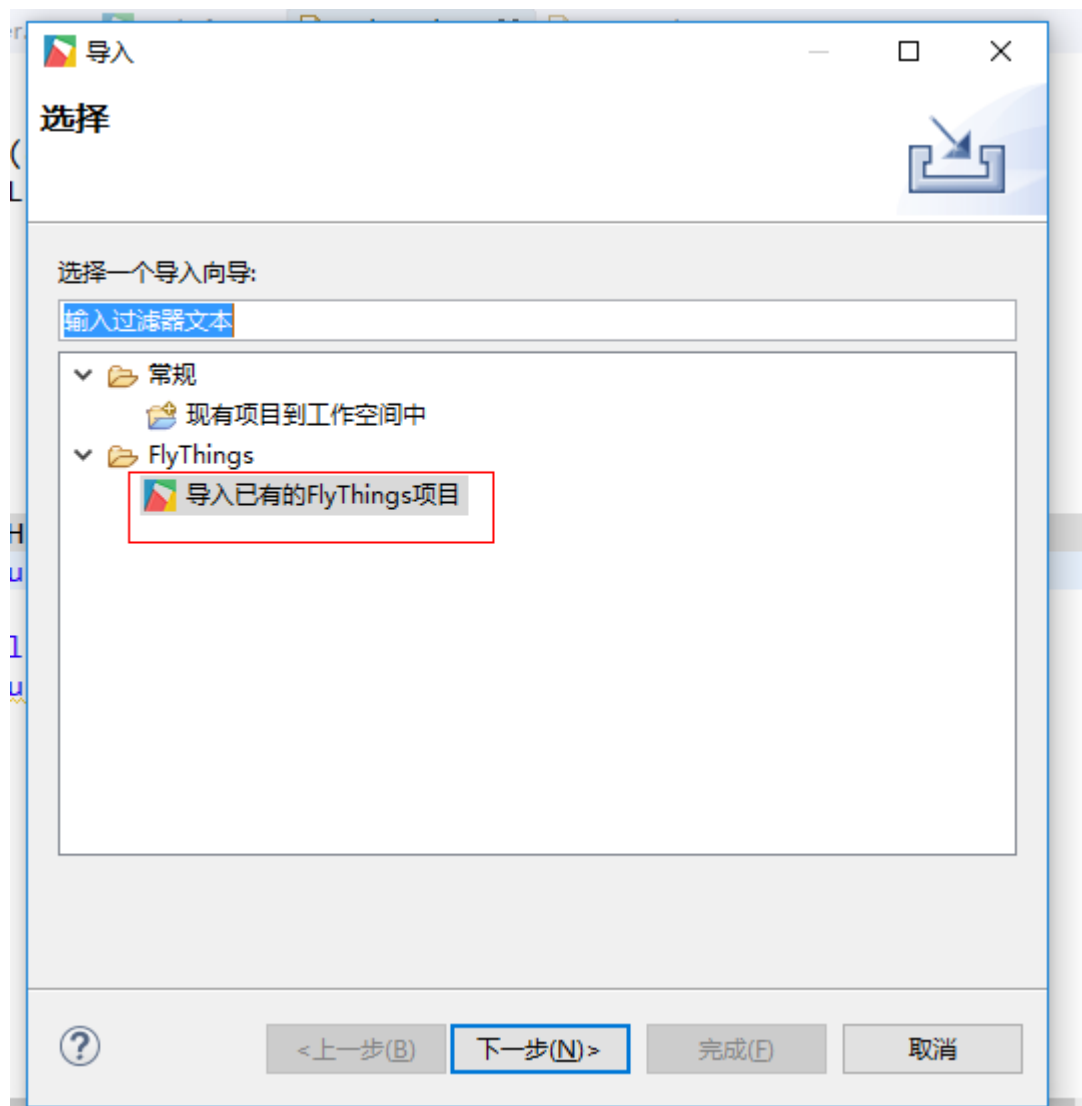
远程站点: /20190815/flythings	
20190815 - flythings - SDK 20190829	
文件名	文件大小
..	
IDE使用参考.pdf	2,937,600
sigmastar-flythings-ide-win32-win32-x86-setup.exe	244,456,000

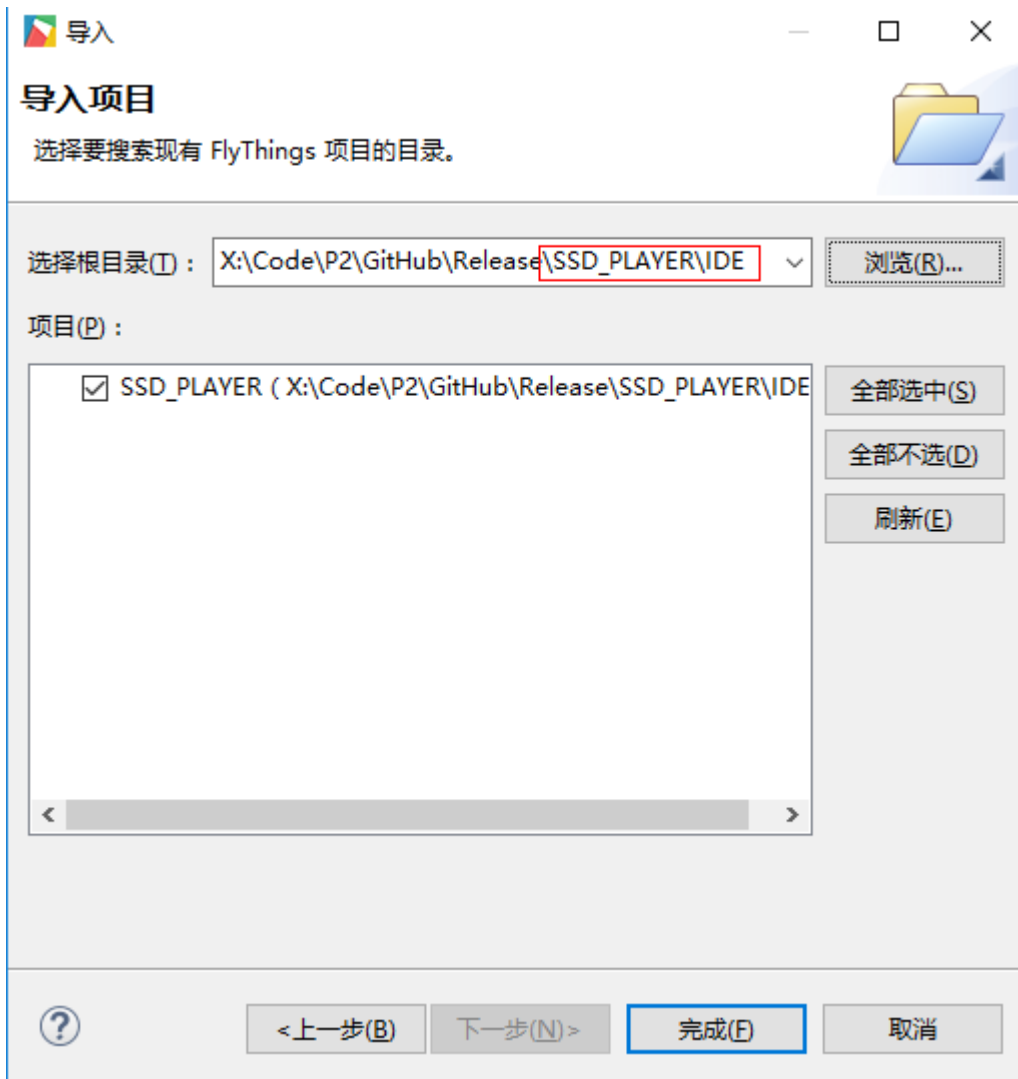
1.2. 2. 导入 SSD_PLAYER 工程

SSD_PLAYER 工程目录放在 FTP 的如下路径:

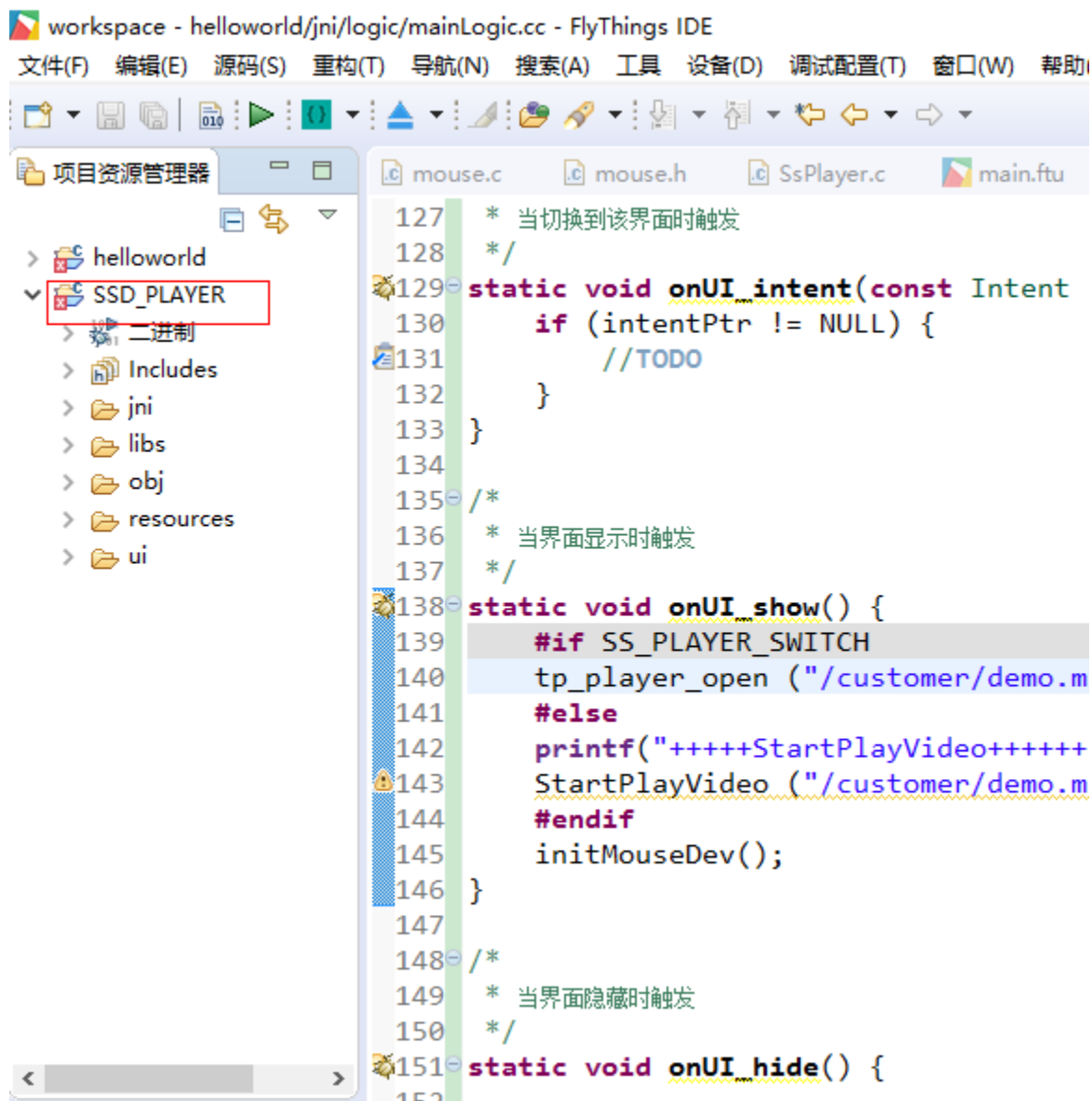
远程站点: /20191029/SDK	
20191029 - DOC - SDK 20191105	
文件名	文件大小
..	
boot-TAKOYAKI-ALPHA005-1024.tar.gz	27,274,000
kernel-TAKOYAKI-ALPHA005-1024.tar.gz	625,694,000
project-TAKOYAKI-ALPHA005-1024.tar.gz	575,571,000
sdk-TAKOYAKI-ALPHA005-1024.tar.gz	97,694,000
spinand_128M_1024x600_image.rar	27,899,000
SSD_PLAYER.tar.gz	295,682,000

解压后，选择 flythings 文件-导入功能，将 SSD_PLAYER\IDE 导入工程:



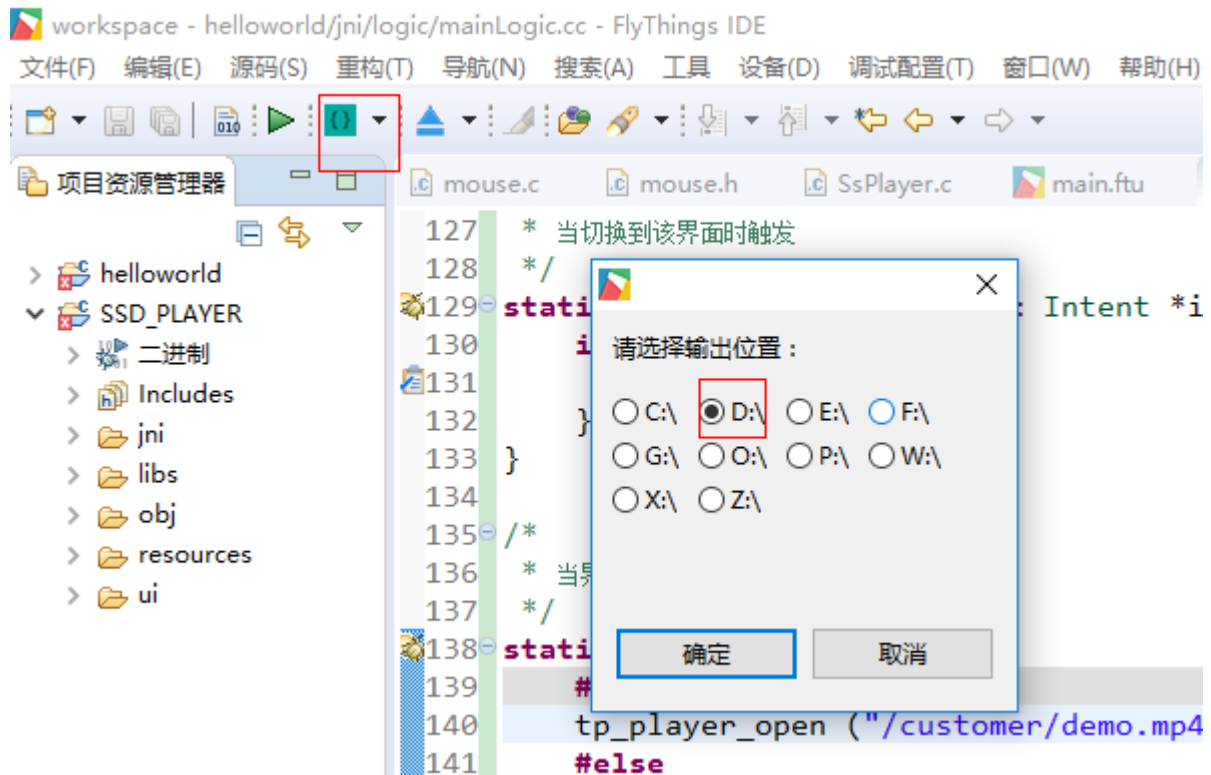


导入后，可以在 flythings 左边的工程栏看到对应的工程：



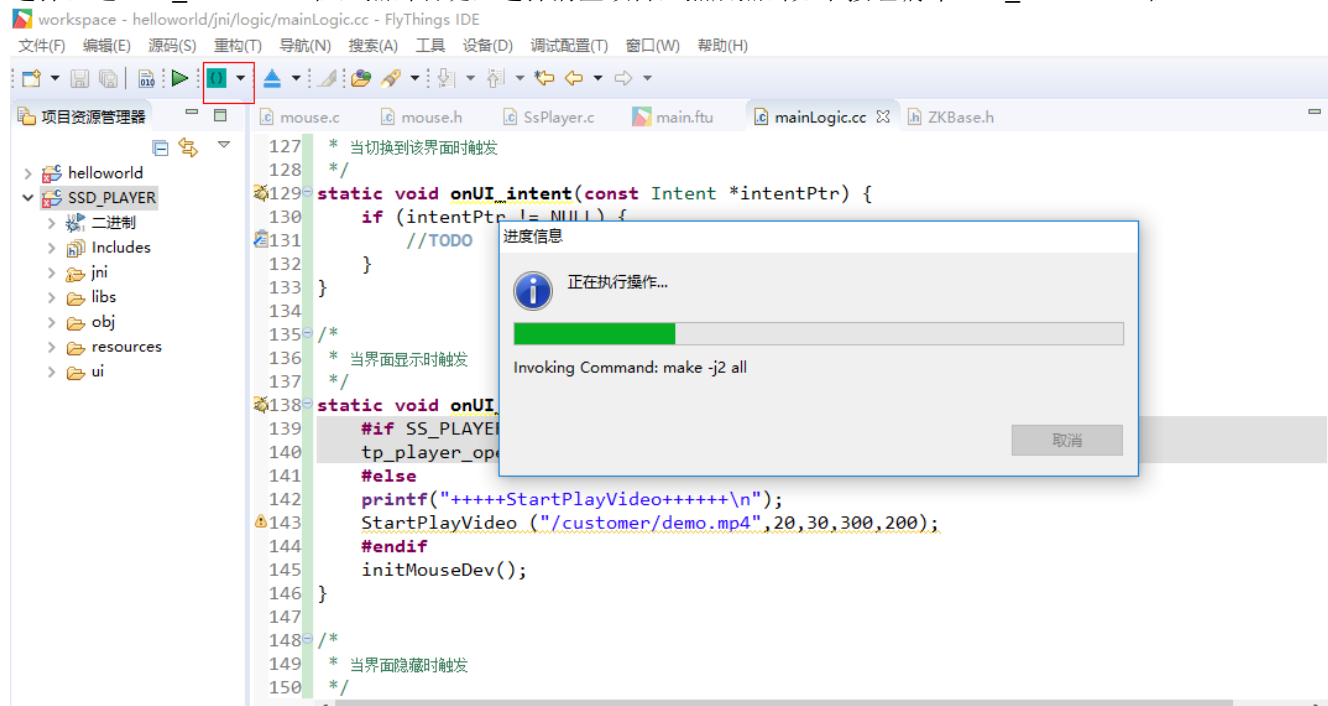
1.3. 设置编译 SSD_PLAYER 生成资源&lib 导出文件路径

设置编译生成的资源和 lib 导出文件路径，目前路径只能存在对应盘符的根目录：



1.4. 编译 SSD_PLAYER 工程

选择左边 SSD_PLAYER 工程，点击右键，选择清空项目，然后点击如下按钮编译 SSD_PLAYER 工程：



编译完成后会在第二步设定的目录下生成 lib 和 res 文件：

此电脑 > 软件 (D:) >

名称	修改日期	类型	大小
EasyUI.cfg	2019/11/7 12:00	CFG 文件	1 KB
udmadedbug.log	2019/10/29 17:44	文本文档	0 KB
zkrestart	2019/10/15 21:21	文件	0 KB
TV_CAM_设备_20190809_110232.mov	2019/8/9 11:02	MOV 文件	6,233 KB
email	2019/11/7 12:08	文件夹	
lib	2019/11/7 12:00	文件夹	
ui	2019/11/6 18:06	文件夹	
font	2019/10/21 12:25	文件夹	
BaiduNetdiskDownload	2019/9/20 10:29	文件夹	

将 lib 和 ui 文件覆盖到 sdk-TAKOYAKI-ALPHA005-1024.tar.gz 解压后的
sdk\verify\application\customer_zk\lib 和 sdk\verify\application\customer_zk\res\ui 目录。

1.5. 编译 project & sdk & kernel & uboot 生成 image

a. 打开 project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk 中的 verify_zk_mini，将
customer_zk_mini app 打包进 customer 分区，并开机自动运行：

```

1 include $(PROJ_ROOT)/release/customer_tailor/nvr_default.mk
2 #interface_ai:=disable
3 #interface_ao:=disable
4 #interface_gfx:=disable
5 interface_hdmi:=disable
6 #interface_divp:=disable
7 #interface_disp:=disable
8 #interface_panel:=disable
9 interface_rgn:=disable
10 interface_shadow:=disable
11 interface_uac:=disable
12 interface_vdf:=disable
13 interface_vdisp:=disable
14 #interface_vdec:=disable
15 interface_venc:=enable
16 interface_wlan:=enable
17
18 misc_fbdev:=enable
19 verify_zk_mini:=enable
20 #mhal
21 #mhal_ao:=disable
22 mhal_csi:=disable
23 #mhal_disp:=disable

```

b. 将 FTP 的/20191029/SDK/Release_to_customer.sh 的编译脚本拷贝到 project 同级目录：

```

aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$ ls -la
total 1267324
drwxr-xr-x  6 aaron.feng sigmastar_users  4096 Nov  7 11:52 .
drwxr-xr-x  7 aaron.feng sigmastar_users  4096 Oct 25 17:07 ..
drwxr-xr-x 21 aaron.feng sigmastar_users  4096 Oct 21 18:42 boot
-rw-r--r--  1 aaron.feng sigmastar_users 15800408 Oct 25 17:05 boot-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x 26 aaron.feng sigmastar_users  4096 Nov  6 20:26 kernel
-rw-r--r--  1 aaron.feng sigmastar_users 625576086 Oct 25 17:05 kernel-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x  8 aaron.feng sigmastar_users  4096 Oct 25 17:04 project
-rw-r--r--  1 aaron.feng sigmastar_users 558616668 Oct 25 17:06 project-TAKOYAKI-ALPHA005-1024.tar.gz
-rwxrwx-rw-  1 aaron.feng sigmastar_users  2829 Nov  7 10:33 Release_to_customer.sh
drwxr-xr-x  4 aaron.feng sigmastar_users  4096 Oct 25 17:04 sdk
-rw-r--r--  1 aaron.feng sigmastar_users 97694764 Nov  6 20:49 sdk-TAKOYAKI-ALPHA005-1024.tar.gz

```

c. 根据 flash type 和芯片型号运行 Release_to_customer.sh 编译 project 生成 image 文件：

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecodes$
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecodes$
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecodes$ ./Release_to_customer.sh -p ssd202 -f nand
chose nand
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
SHIPPED scripts/kconfig/zconf.tab.c
SHIPPED scripts/kconfig/zconf.lex.c
SHIPPED scripts/kconfig/zconf.hash.c
HOSTCC scripts/kconfig/zconf.tab.o
HOSTLD scripts/kconfig/conf
#
# configuration written to .config
```

编译完成后，会生成 image 目录，然后通过 tftp 烧录即可：

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecodes$ ls -la
total 1267328
drwxr-xr-x 7 aaron.feng sigmastar_users 4096 Nov 7 12:22 .
drwxr-xr-x 7 aaron.feng sigmastar_users 4096 Oct 25 17:07 ..
drwxr-xr-x 21 aaron.feng sigmastar_users 4096 Nov 7 12:20 boot
-rw-r--r-- 1 aaron.feng sigmastar_users 15800408 Oct 25 17:05 boot-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x 4 aaron.feng sigmastar_users 4096 Nov 7 12:22 images
drwxr-xr-x 26 aaron.feng sigmastar_users 4096 Nov 7 12:22 kernel
-rw-r--r-- 1 aaron.feng sigmastar_users 625576086 Oct 25 17:05 kernel-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x 8 aaron.feng sigmastar_users 4096 Oct 25 17:04 project
-rw-r--r-- 1 aaron.feng sigmastar_users 558616668 Oct 25 17:06 project-TAKOYAKI-ALPHA005-1024.tar.gz
-rwxrwx-rw- 1 aaron.feng sigmastar_users 2676 Nov 7 12:04 Release_to_customer.sh
drwxr-xr-x 4 aaron.feng sigmastar_users 4096 Oct 25 17:04 sdk
-rw-r--r-- 1 aaron.feng sigmastar_users 97694764 Nov 6 20:49 sdk-TAKOYAKI-ALPHA005-1024.tar.gz
```

默认编译的 SSD_PLAYER 工程不带 wifi 和播放器功能，如需要打开，需按如下方法打开。

全功能 FLYTHINGS IDE 环境搭建

全功能 flythings 占用的 flash 空间和内存都比较大，需要 **ssd202 & spinand 128M** 的配置才可以跑起来，具体配置如下：

1.6. 增大 mma heap 内存配置

Nor flash:

将 project\configs\nvr\i2m\8.2.1\nor.glibc-squashfs.011a.128 如下位置改成 0x3800000:

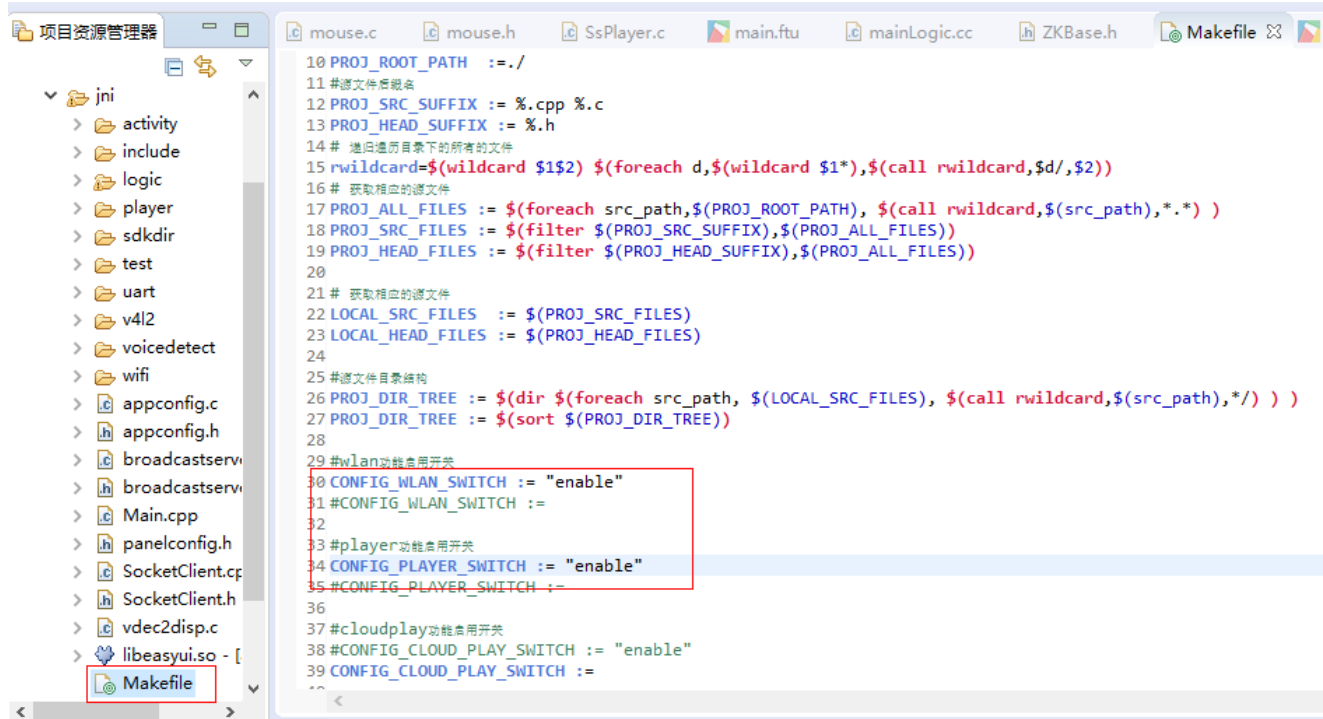
```
CHIP = i2m
BOARD = 011A
BOARD_NAME = SSC011A-S01A
PRODUCT = nvr
TOOLCHAIN = glibc
TOOLCHAIN_VERSION = 8.2.1
KERNEL_VERSION = 4.9.84
LIBC = libc-2.28
BUSYBOX = busybox-1.20.2-arm-linux-gnueabi-hf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = nor.squashfs.hfglibc.nvr.mma
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_128M.h
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x7D00000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070CP50
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLN) mma_heap=mma_heap_name0,miu=0,sz=0x3800000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabi-hf-
```

Nand flash:

将 project\configs\nvr\i2m\8.2.1\spinand.glibc.011a.128 的如下位置改成 0x3800000:

```
CHIP = i2m
BOARD = 011A
BOARD_NAME = SSC011A-S01A
PRODUCT = nvr
TOOLCHAIN = glibc
TOOLCHAIN_VERSION = 8.2.1
KERNEL_VERSION = 4.9.84
LIBC = libc-2.28
BUSYBOX = busybox-1.20.2-arm-linux-gnueabi-hf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = spinand.ubifs.hfglibc.p2.mma
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_128M.h
MHAL = i2m
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x7D00000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070BO30I21Y0
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLN) mma_heap=mma_heap_name0,miu=0,sz=0x3800000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabi-hf-
```

1.7. 打开 SSD_PLAYER wifi & player 选项:



剩下编译步骤跟精简版 flythings IDE 环境搭建一致。

FLYTHINGS UI 调整触摸屏触控分辨率

触摸屏的触控分辨率需要保持跟 framebuffer 的分辨率一致，framebuf 的分辨率参考板子/config/fbdev.ini:

```
/ # cat /config/fbdev.ini
[FB_DEVICE]
FB_HWLAYER_ID = 1
FB_HWIN_ID = 0
FB_HWLAYER_DST = 3
FB_HWIN_FORMAT = 5
FB_HWLAYER_OUTPUTCOLOR = 1
FB_WIDTH = 1024
FB_HEIGHT = 600
FB_TIMMING_WIDTH = 1920
FB_TIMMING_HEIGHT = 1080
FB_MMAP_NAME = E_MMAP_ID_FB
FB_BUFFER_LEN = 8192
#unit:Kbyte,4096=4M, fbdev.ko alloc size = FB_BUFFER_L
```

配置方法如下(1024x600.bin & 800x480.bin 放在 ftp: /20191029/SDK/SSD_PLAYER.tar.gz):

1024x600:

echo 1024x600.bin > /sys/bus/i2c/devices/1-005d/gtcfg

800x480:

echo 800x480.bin > /sys/bus/i2c/devices/1-005d/gtcfg

FLYTHINGS UI 调整 PANEL 分辨率

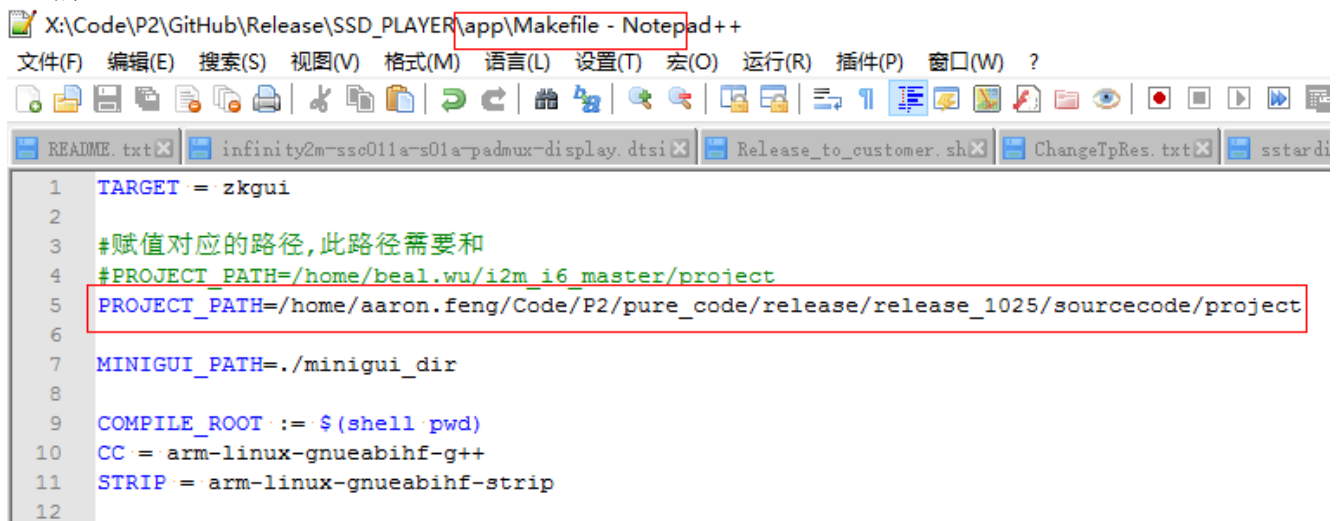
目前公版 flythings UI 主要适配 1024x600 和 800x480 两种分辨率的 panel, 默认的配置是 1024x600 的分辨率, 如果要改成 800x480 的分辨率, 配置方法如下:

1.8. 屏参修改成 800x480

将 FTP/20191029/SDK/SSD_PLAYER.tar.gz 解压, 将 SSD_PLAYER\app\sstardisp.c 的 UI_1024_600 置为 0:

```
1  #include "string.h"
2  #include "stdio.h"
3  #include "stdlib.h"
4
5  #include "mi_sys.h"
6  #include "sstardisp.h"
7
8  #include "mi_panel_datatype.h"
9  #include "mi_panel.h"
10 #include "mi_disp_datatype.h"
11 #include "mi_disp.h"
12
13
14 #define UI_1024_600 0
15 #define USE_MIPI 0
16
17
18 #if UI_1024_600
19 #include "SAT070CP50_1024x600.h"
20 #else
21 #if USE_MIPI
22 #include "ST7701S_480x480_MIPI.h"
23 #else
24 #include "SAT070AT50_800x480.h"
25 #endif
26 #endif
```

进入到 SSD_PLAYER\app, 重新编译 zkgui, 编译前需要将 app 下面的 makefile project 路径改成你实际的 project 路径:



```
1  TARGET = zkgui
2
3  #赋值对应的路径,此路径需要和
4  #PROJECT_PATH=/home/beal.wu/i2m_i6_master/project
5  PROJECT_PATH=/home/aaron.feng/Code/P2/pure_code/release/release_1025/sourcecode/project
6
7  MINIGUI_PATH=./minigui_dir
8
9  COMPILE_ROOT := $(shell pwd)
10 CC = arm-linux-gnueabi-g++
11 STRIP = arm-linux-gnueabi-strip
12
```

然后直接 make clean;make 即可:



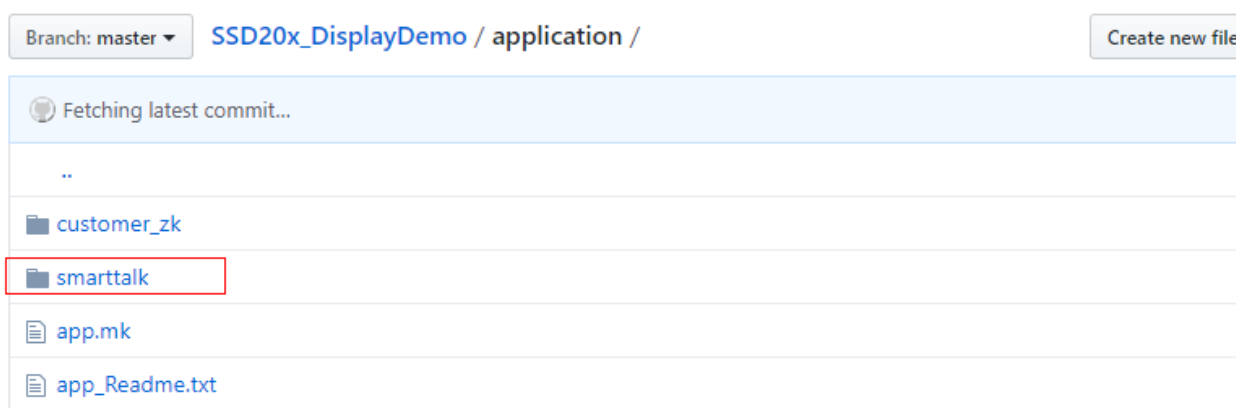
编译完将 SSD_PLAYER\app\bin\zkgui 替换到 sdk\verify\application\customer_zk_xx\bin 下面，重名为 zkgui_800_480，然后执行 1.5 步骤，重新编译烧录即可；

2. MINIGUI SMARTTALK 环境搭建

smarttalk 比较大, nor 放不下必须要 spinand 才能跑, demo 放在:

https://github.com/wuyanxiong/SSD20x_DisplayDemo/tree/master/application

将 smarttalk 拷贝到 sdk\verify\application, 再按如下步骤配置即可。



2.1. Sdk 打开 smarttalk 编译选项

project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk(smarttalk 和 flythings 不能共存, 打开 verify_smarttalk 需要屏蔽 verify_zk_mini):

```
1 include $(PROJ_ROOT)/release/customer_tailor/nvr_default.mk
2 #interface_ai:=disable
3 #interface_ao:=disable
4 #interface_gfx:=disable
5 interface_hdmi:=disable
6 #interface_divp:=disable
7 #interface_disp:=disable
8 #interface_panel:=disable
9 interface_rgn:=disable
10 interface_shadow:=disable
11 interface_uac:=disable
12 interface_vdf:=disable
13 interface_vdisp:=disable
14 #interface_vdec:=disable
15 interface_venc:=enable
16 interface_wlan:=enable
17
18 misc_fbdev:=enable
19 #verify_zk_mini:=enable
20 verify_smarttalk:=enable
21 #mhal
22 #mhal_aio:=disable
23 mhal_csi:=disable
24 #mhal_disp:=disable
```

2.2. 编译 project & sdk & kernel

Smarttalk 播放视频比较耗内存, 编译前需要将对应 config 的 mma 的 size 加大到 0x2000000:

```
*X:\Code\P2\pure_code\release\release_1025\sourcecode\project\configs\nvr\i2m\8.2.1\spinand.glibc.011a.64 - Notepad++
文件(F) 编辑(E) 搜索(S) 视图(V) 格式(M) 语言(L) 设置(T) 宏(O) 运行(R) 插件(P) 窗口(W) ?
Release_to_customer.sh3 ChangeTpRes.txt sstardisp.o Makefile image.mk app.mk nvr_i2m_display_glibc_tailor.mk
1 CHIP = i2m
2 BOARD = 011A
3 BOARD_NAME = SSC011A-S01A
4 PRODUCT = nvr
5 TOOLCHAIN = glibc
6 TOOLCHAIN_VERSION = 8.2.1
7 KERNEL_VERSION = 4.9.84
8 LIBC = libc-2.28
9 BUSYBOX = busybox-1.20.2-arm-linux-gnueabi-hf-glibc-8.2.1-dynamic
10 KERNEL_CONFIG = glibc
11 IMAGE_CONFIG = spinand.ubifs.hfglibc.p2.mma
12 CUSTOMER_OPTIONS = 011a.201_options.mk
13 CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
14 MMAP = MMAP_I2M_64M.h
15 MHAL = i2m
16 BOOTLOGO_FILE = sigmastar1024_600.jpg
17 #BOOTLOGO_ADDR = E_LX_FB
18 LOGO_ADDR = 0x3D00000
19 BOOTLOGO_BUFSIZE = 0x200000
20 PANEL_NAME = SAT070B030I21Y0
21 PANEL_WIDTH = 1024
22 PANEL_HEIGHT = 600
23 EXBOOTARGS =
24 KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLEN) mma_heap=mma_heap_name0,miu=0,sz=0x2000000 mma_memblock_remove=1
25 TOOLCHAIN_REL = arm-linux-gnueabi-hf-
26
```

编译烧录步骤跟 1.5 一致

2.3. 调整屏幕分辨率

公版目前主要适配了 1024x600 和 800x480 两种分辨率的 panel，默认配置是 800x480，如果需要改成 1024x600 配置方法如下：

2.3.1 修改屏参文件

sdk\verify\application\smartrtalk\demoApp\main_ui.h 打开 UI_1024_600:

```
7
8 #define UI_ARGB888_FF UI_ARGB2PIXEL8888(255,255,255,255)
9
10 #define UI_ARGB888_KEY UI_ARGB2PIXEL8888(128,128,128,128)
11
12
13 #define UI_DISP_W .....1920
14 #define UI_DISP_H .....1080
15
16 #define UI_1024_600
17
18 #ifdef UI_1024_600
19 #define MAINWND_W .....1024
20 #define MAINWND_H .....600
21 #else
22 #define MAINWND_W .....800
23 #define MAINWND_H .....480
24 #endif
```

重新 build sdk\verify\application\smartrtalk\demoApp:

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/verify/application/smartrtalk/demoApp$ rm -rf build;./build_sample.sh
~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/verify/application/smartrtalk/demoApp/build ~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/ver
-- The C compiler identification is GNU 5.4.0
-- The CXX compiler identification is GNU 5.4.0
-- Check for working C compiler: /usr/bin/cc
-- Check for working C compiler: /usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
```

2.3.2 修改 framebuffer 宽高和大小

project\board\i2m\SSC011A-S01A\config\fbdev.ini:

```

1  [FB_DEVICE]
2  FB_HWLAYER_ID = 1
3  FB_HWWIN_ID = 0
4  FB_HWLAYER_DST = 3
5  FB_HWWIN_FORMAT = 5
6  FB_HWLAYER_OUTPUTCOLOR = 1
7  FB_WIDTH = 1024
8  FB_HEIGHT = 600
9  FB_TIMMING_WIDTH = 1920
10 FB_TIMMING_HEIGHT = 1080
11 FB_MMAP_NAME = E_MMAP_ID_FB
12 FB_BUFFER_LEN = 8196
13 #unit:Kbyte, 4096=4M, fbdev.ko alloc size = FB_BUFFER_LEN*1024
14

```

2.3.3 修改 MiniGUI.cfg 相关配置

sdk\verify\application\smarstalk\RunFile\etc\MiniGUI.cfg

```

8
9  [system]
10 # GAL engine and default options
11 #gal_engine=fbcon
12 gal_engine=sstar
13 #defaultmode=800x480-32bpp
14 mmaBufSize=0xF00000
15 defaultmode=1024x600-32bpp
16
17 # IAL engine
18 #ial_engine=console
19 ial_engine=tslib
20 #mdev=/dev/input/mice
21 mdev=/dev/input/event0
22 mtype=IMPS2
23 #mtype=none
24
25 [sstar]
26 #defaultmode=800x480-32bpp
27 defaultmode=1024x600-32bpp
28

```

2.3.4 修改 mini gui 布局文件

sdk\verify\application\smarstalk\image.mk:



```
app:
ifeq ($(IMAGE_INSTALL_DIR),)
    →@echo "directory of image is not defined"
    →@exit 1
endif
→@cp $(APPLICATION_PATH)/smarstalk/demoApp/build/mginit $(IMAGE_INSTALL_DIR)/customer
→@$(TOOLCHAIN_REL)strip --strip-unneeded $(IMAGE_INSTALL_DIR)/customer/mginit
→@cp -rf $(APPLICATION_PATH)/smarstalk/RunFile/appres $(IMAGE_INSTALL_DIR)/customer
→@cp -rf $(APPLICATION_PATH)/smarstalk/RunFile/CSpotter $(IMAGE_INSTALL_DIR)/customer
→@cp -rf $(APPLICATION_PATH)/smarstalk/RunFile/minigui $(IMAGE_INSTALL_DIR)/customer
→@cp -rf $(APPLICATION_PATH)/smarstalk/RunFile/lib $(IMAGE_INSTALL_DIR)/customer
→@cp -rf $(APPLICATION_PATH)/smarstalk/RunFile/etc/* $(IMAGE_INSTALL_DIR)/rootfs/etc/
→@cat $(APPLICATION_PATH)/smarstalk/RunFile/demo.sh >> $(IMAGE_INSTALL_DIR)/customer/demo.sh
→@cp $(APPLICATION_PATH)/smarstalk/RunFile/hscrollview.rc $(IMAGE_INSTALL_DIR)/customer
→@cp $(APPLICATION_PATH)/smarstalk/RunFile/*.xml $(IMAGE_INSTALL_DIR)/customer
→@cp $(APPLICATION_PATH)/smarstalk/RunFile/720P25.h264 $(IMAGE_INSTALL_DIR)/customer
→@cp $(APPLICATION_PATH)/smarstalk/RunFile/8K_16bit_MONO.wav $(IMAGE_INSTALL_DIR)/customer
→#@mv $(IMAGE_INSTALL_DIR)/customer/smartLayout_800x480.xml $(IMAGE_INSTALL_DIR)/customer/smartLayout.xml
→@mv $(IMAGE_INSTALL_DIR)/customer/smartLayout_1024x600.xml $(IMAGE_INSTALL_DIR)/customer/smartLayout.xml
```